Laplace Transform

Definition: **Forward transform:** $$F(s) = \mathcal{L}\big(f(t)\big) = \int_0^{\infty} f(t)e^{-st}dt$$

Inverse transform: $$f(t) = \mathcal{L}^{-1}\big(F(s)\big) = \frac{1}{2\pi j}\int_{\gamma-j\infty}^{\gamma+j\infty} F(s)e^{st}d\omega$$

Properties: (α, β = constant)

1) Linearity:	$\alpha f(t) + \beta g(t) \overset{\mathcal{L}}{\leftrightarrow} \alpha F(s) + \beta G(s)$
2) Scaling:	$f(\alpha t) \overset{\mathcal{L}}{\leftrightarrow} \frac{1}{\alpha} F\left(\frac{s}{\alpha}\right)$
3) Second translation:	$f(t-\alpha)u(t-\alpha) \overset{\mathcal{L}}{\leftrightarrow} e^{-s\alpha}F(\omega); \alpha > 0$
4) First translation:	$e^{t\alpha}f(t) \overset{\mathcal{L}}{\leftrightarrow} F(s-\alpha)$
5) Time derivative:	$\frac{df(t)}{dt} \overset{\mathcal{L}}{\leftrightarrow} sF(s) - f(0)$
NOTE: $f^{(n)}(t) \overset{\mathcal{L}}{\leftrightarrow} s^nF(s) - s^{n-1}f(0) - s^{n-2}f'(0) - s^{n-3}f''(0) \ldots$	
6) Multiplication by *t*:	$-tf(t) \overset{\mathcal{L}}{\leftrightarrow} \frac{dF(s)}{ds}$
NOTE: $t^nf(t) \overset{\mathcal{L}}{\leftrightarrow} (-1)^n \frac{d^nF(s)}{ds^n}$	
7) Time integration	$\int_0^t f(\tau)d\tau \overset{\mathcal{L}}{\leftrightarrow} \frac{F(s)}{s}$
8) Division by *t*:	$\frac{f(t)}{t} \overset{\mathcal{L}}{\leftrightarrow} \int_s^{\infty} F(u)du$
9) Time convolution:	$f * g(t) = \int_0^t f(\tau)g(t-\tau)d\tau \overset{\mathcal{L}}{\leftrightarrow} F(s)G(s)$
10) Periodic function:	$F_{periodic}(t) \overset{\mathcal{L}}{\leftrightarrow} \frac{F_{1period}(s)}{1-e^{-sT}}$
11) Initial value problem:	$\lim_{t\to 0} f(t) = \lim_{s\to\infty} sF(s)$
12) Final value problem	$\lim_{t\to\infty} f(t) = \lim_{s\to 0} sF(s)$

Laplace transform of common functions:

f(t)	F(s)
$\delta(t)$	1
$u(t), 1$	$\frac{1}{s}; s > 0$
t^n	$\frac{n!}{s^{n+1}}; s > 0$
$e^{\alpha t}$	$\frac{1}{s-\alpha}; s > \alpha$
$\cos(\alpha t)$	$\frac{s}{s^2+\alpha^2}; s > 0$
$\sin(\alpha t)$	$\frac{\alpha}{s^2+\alpha^2}; s > 0$
$\cosh(\alpha t)$	$\frac{s}{s^2-\alpha^2}; s > \lvert\alpha\rvert$
$\sinh(\alpha t)$	$\frac{\alpha}{s^2-\alpha^2}; s > \lvert\alpha\rvert$